FRAUD PREVENTION & DETECTION TECHNOLOGIES

Relative Size Tests using Excel

GROUP 12

School of Applied Sciences and Education

ICT Department

Cyber Security

STUDENT NAME	STUDENT NUMBER
James Soko	20231392
Cuthbelt Ndhlovu	20231784
Courage Silweya	20230099
Esther Ngoma	20230442
Alice M. Chota	22175367
Alinet Hang'oba	20231685

QUESTION

- Objective: Compare item sizes to benchmarks (e.g., transactions > 3x average).

- **Deliverables:**

a) Z-score/percentile analysis.

b) A report on "unusually large" items.

- Example: Flag Ponzi scheme pay-outs (e.g., 10x typical returns).

OBJECTIVES

- Understanding the term **data set**.
- Understanding the term **mean** and how to calculate it.
- Understanding **Standard Deviation** and how to use the formula.
- Understanding **Zero-score** and how to calculate the Standard Score.
- Understanding how to integrate the data set and use **AVERAGE**, **STDEV**, **PECENTRANK.INC**, **IF**, and **COUNTIF** functions in **Excel**.
- Understanding how to calculate Zero-score for each transaction in **Excel**.
- Understanding how to use the above tools or methods in **Fraud Prevention & Detection Technologies**.

DATA SET

- A **data set** is a collection of related data, often organized in a structured format like a table or spreadsheet, used for analysis, model training, or other purposes.

Transaction ID	Date	Account	Amount (K)	Name	Z-score	Pencentile Rank	Status
TXN001	07 May 2025	7817080100846	750.00	James Soko	-0.05	80%	Usual
TXN002	07 May 2025	7817080100761	325.00	Cuthbert Ndlovu	-0.48	40%	Usual
TXN003	07 May 2025	7817080100333	2,800.00	Courage Silweya	2.01	100%	Unusual
TXN004	07 May 2025	7817080100909	425.00	Esther Ngoma	-0.38	60%	Usual
TXN005	07 May 2025	7817080100123	300.00	Alice M. Chota	-0.51	20%	Usual
TXN006	07 May 2025	7817080100000	225.00	Alinet Hang'oba	-0.58	0%	Usual

MEAN

- **Mean** is the average of the given numbers.
- The formula for the Mean is $$m = \frac{\text{sum of the terms}}{\text{number of terms}}$$

μ = 750+325+2800+425+300+225 ÷ 6

= 4825 ÷ 6

= 804.16

The mean (m) for the **data set** given is 804.16 for the transactions.

STANDARD DEVIATION

- **Standard Deviation** is a quantity expressing by how much the members of a group differ from the mean value for the group.
- The formula for the Standard Deviation is

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

- σ = dataset standard deviation
- N = the size of the dataset
- Xí = each value from the dataset
- μ = the dataset mean (m)

STANDARD DEVIATION

Formula

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$

μ = 750+325+2800+425+300+225 ÷ 6

= 4825 ÷ 6

= 804.16

N = 6

Sub-standard Deviations

750-804.16 = -54.16

325-804.16 = -479.16

2800-804.16 = 1995.84

425-804.16 = -379.16

300-804.16 = -504.16

225-804.16 = -579.16

STANDARD DEVIATION

Sub-standard Deviations²

$-54.16^2 = 2933.30$

$-479.16^2 = 229594.30$

$1995.84^2 = 3983377.30$

$-379.16^2 = 143762.30$

$-504.16^2 = 254177.30$

$-579.16^2 = 335426.30$

Summation

$\sum = (2933.30 + 229594.30 + 3983377.30 + 143762.30 + 254177.30 + 335426.30)$

$\sum = 4949270.8$

$\sum \div N = 824878.46$

$\checkmark\ 824878.46 = 908.22$

STANDARD SCORE

- The **Z-score** measures how many standard deviations a value is from the mean.
- The formula for the Standard Score or Z-score is $$Z = \frac{x - \mu}{\sigma}$$
- Z = standard score
- X = observed value
- μ = mean of the dataset (m)
- σ = standard deviation of the dataset

STANDARD SCORE

Formula

$$Z = \frac{x - \mu}{\sigma}$$

= 750-804.16 ÷ 908.22

= -54.16÷908.22

= -0.05

Therefore -0.05 is the Standard Score for James's transaction (K750)

Now let's use **Excel** to find all the Zero-score values as shown below

Amount (K)	Name	Z-score
750.00	James Soko	-0.5
325.00	Cuthbert Ndlovu	-0.48
2800.00	Courage Silweya	2.01
425.00	Esther Ngoma	-0.38
300.00	Alice M. Chota	-0.51
225.00	Alinet Han'goba	-0.58

PERCENTILE RANK

- A **percentile score** tells you the relative standing of a value in a dataset.
- In fraud detection, very **high percentile scores** (e.g., above the 95th, 98th, or 99th percentile) often signal **unusually large** or **suspicious transactions**, which may warrant investigation.

Percentile Rank	Interpretation	Action
< 90	Normal Range	No action
90 – 95	Mildly Unusual	Flag for review
> 95	Unusual (possible red flag)	Investigate for fraud
> 99	Very rare (likely suspicious)	Investigate for high fraud risk

EXCEL

- **Excel** is a spreadsheet program from Microsoft used for organizing, analyzing, and visualizing data.
- Download the excel lab **here**.
- Formulas used:

a) **Mean** =AVERAGE(D5:D10)

b) **Standard** Deviation =STDEV.S(D5:D10)

c) **Z-score** =(D5 - $K$5)/($K$6)

d) **Pecentile Rank** =PERCENTRANK.INC($D$5:$D$10, D5)

EXCEL

e) **Filter** =FILTER(A5:H10,H5:H10=D14,"No search as been made or zero results")

f) **Count if**

i. =COUNTIF(H5:H10,"Usual")

ii. =COUNTIF(H5:H10,"Unusual")

EXCEL LAB: RELATIVE SIZE TESTS FOR FRAUD DETECTION

- Analyzing transaction data to flag potentially fraudulent activity using Z-scores and relative size thresholds (e.g., Z-score > 2 or 2× average).

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HELSB MEAL ALLOWANCES FOR GROUP 12

Transaction ID	Date	Account	Amount (K)	Name	Z-score	Pencentile Rank	Status
TXN001	07 May 2025	7817080100846	750.00	James Soko	-0.05	80%	Usual
TXN002	07 May 2025	7817080100761	325.00	Cuthbert Ndlovu	-0.48	40%	Usual
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Summary Statistics	
Mean (Average)	804.17
Standard Dev.	994.91
Usual	5
Unusual	1

			Results						
	Search by Usual or Unusual:		Usual	Date	Account	Amount (K)	Name	Z-score	Pencentile Rank	Status
Action	Confirm	Edit	TXN001	07 May 2025	7817080100846	750	James Soko	-0.05	80%	Usual
	Confirm	Edit	TXN002	07 May 2025	7817080100761	325	Cuthbert Ndlovu	-0.48	40%	Usual
	Confirm	Edit	TXN004	07 May 2025	7817080100909	425	Esther Ngoma	-0.38	60%	Usual
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FRAUD DETECTION & PREVENTION TECHNOLOGIES

- **Fraud detection technologies** include tools, methods, and systems used to identify suspicious or unauthorized activities (like money laundering, embezzlement, Ponzi schemes, or invoice fraud).
- These may include: **Data analytics tools** (like ACL, IDEA, **Excel**)
- **Relative Size Test (RST)** as a Detection Tool helps detect outliers by flagging transactions that are:

a) Statistically unusual (e.g., Z-score > 2)

b) Disproportionately large compared to peer values (e.g., >2× average)

c) Beyond a defined percentile threshold (e.g., >99th percentile)